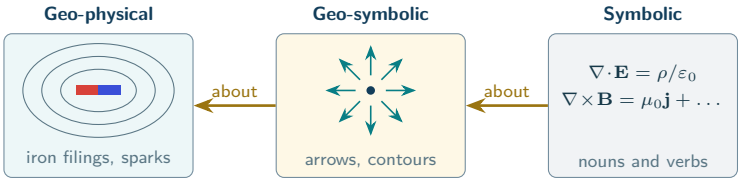
PHYSICS ON ONE PAGE

What does an equation represent?

Maxwell's equations as sentences about patterns, built up from Coulomb's law and the magnetic force law.

1 / THE RULE AND THE PICTURE

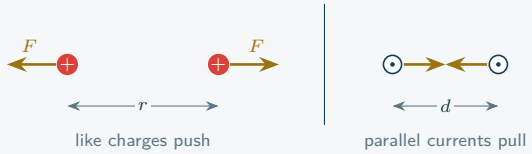
An equation is about a pattern, in the sense of Brentano's aboutness



The filings, sparks and compass needle are a physical pattern; the arrows and contours are a picture of it; the equation says in symbols what the picture does. Its nouns ($\mathbf{E}$, $\mathbf{B}$, ρ, $\mathbf{j}$) name the things and its verbs (spreading, swirling, sloping, changing) name what is done to them.

2 / FORCE LAWS / STATIC CHARGES AND STEADY CURRENTS

Coulomb's law and the wire-to-wire force, then the fields they imply



Coulomb's law is $F = kQq/r^2$ with $k = 1/(4\pi\varepsilon_0) = 8.99\times10^9$, so $1.0\ \mu\mathrm{C}$ and $2.0\ \mu\mathrm{C}$ at $0.30\ \mathrm{m}$ push with $0.20\ \mathrm{N}$. Two long parallel wires obey $F/L = \mu_0 I_1 I_2/(2\pi d)$, so $10\ \mathrm{A}$ in each at $5.0\ \mathrm{cm}$ pull with $4.0\times10^{-4}\ \mathrm{N\,m^{-1}}$, using $\mu_0/2\pi = 2\times10^{-7}$. Dividing out the test object leaves fields, $E = F/q = kQ/r^2$ and $B = \mu_0 I/(2\pi r)$, which exist whether or not anything is there to feel them.

3 / SPREADING / DIVERGENCE, FROM FIELD LINES THROUGH A SPHERE

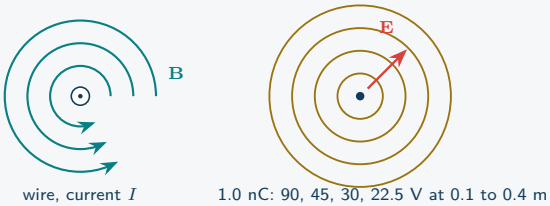
The same lines spread over $4\pi r^2$, which is why Coulomb falls as $1/r^2$



Count the field lines leaving a sphere of radius r round charge Q: the field is the same all over it, so the count is $E\cdot4\pi r^2$, and if that count is Q/ε_0 for every r then $E = Q/(4\pi\varepsilon_0 r^2) = kQ/r^2$. Locally this is divergence, $\nabla\cdot\mathbf{E} = \rho/\varepsilon_0$: field spreads out of a small region in proportion to the charge in it. Spreading arrows read positive, converging arrows negative and parallel arrows zero. Magnets give zero every time, $\nabla\cdot\mathbf{B} = 0$, since each line entering a region also leaves it.

4 / SWIRLING AND SLOPING / CIRCULATION AND THE GRADIENT OF POTENTIAL

Currents make fields swirl, and static fields roll downhill on a potential map

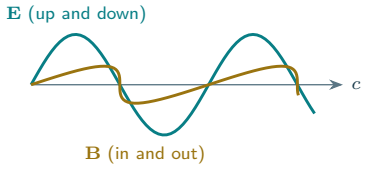


Walking a loop of radius r round a wire and adding the field along the path gives $B\cdot2\pi r = \mu_0 I$, the field of the force law; this circulation is the swirl in $\nabla\times\mathbf{B} = \mu_0\mathbf{j}$. A changing magnetic flux makes an electric swirl too, $\mathcal{E} = -\mathrm{d}\Phi/\mathrm{d}t$, as when a magnet pushed into a coil lights a bulb. Static electric fields do not swirl, so they come from a landscape of potential with $E = -\mathrm{d}V/\mathrm{d}x$: the field points downhill and is strongest where the contours crowd, and for $1.0\ \mathrm{nC}$ the potential $V = 9.0/r$ volts gives $899\ \mathrm{V\,m^{-1}}$ at $0.10\ \mathrm{m}$.

5 / MAXWELL / FOUR SENTENCES ABOUT PATTERNS

Each equation pairs nouns with a verb, and the last two make light

equation	verb	about	in words
$\nabla\cdot\mathbf{E} = \rho/\varepsilon_0$	spreading	static cling, sparks	electric field lines begin and end on charge
$\nabla\cdot\mathbf{B} = 0$	spreading	iron filings, cut magnets	magnetic field lines never begin or end
$\nabla\times\mathbf{E} = -\partial\mathbf{B}/\partial t$	swirl, rate	induction in a coil	a changing $\mathbf{B}$ makes $\mathbf{E}$ swirl
$\nabla\times\mathbf{B} = \mu_0\mathbf{j} + \mu_0\varepsilon_0\,\partial\mathbf{E}/\partial t$	swirl, rate	compass round a wire, light	current and a changing $\mathbf{E}$ make $\mathbf{B}$ swirl



The last two couple the fields, a changing $\mathbf{E}$ making $\mathbf{B}$ and a changing $\mathbf{B}$ making $\mathbf{E}$, so a disturbance sustains itself and travels at

$$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = 2.998\times10^8\ \mathrm{m\,s^{-1}},\ \text{the speed of light}$$

ASIDE / WHAT THE FILINGS ARE

The iron filings are just there, in themselves; the arrows are our map of them, and the equations are about both.

ASIDE / CHECK IN PYTHON

`1/sqrt(mu_0*epsilon_0)` gives 2.998×10^8 and `k*1e-6*2e-6/0.3**2`, with $k = 8.99\times10^9$, gives 0.1997.

Convention. Differential forms shown, with the integral forms used in the working; vacuum with $\mu_0\varepsilon_0 = 1/c^2$. After the slide on geo-physical, geo-symbolic and symbolic levels.